Uncertainty-Aware Selection of PSO-Tuned SVMs

Implementation audit, reproducibility checks, and lambda sensitivity analysis

Purpose
Consolidate the verified state of the repository, describe what the current experiments support, and identify the remaining research tasks. This report distinguishes implemented work from claims that still require validation.

Prepared from the uploaded project archive and its saved experiment outputs.

1. Executive summary

The repository contains a working nested-CV PSO-SVM comparison, a separate inner-OOF threshold-calibration experiment, and a post-hoc lambda sensitivity analysis. The preprocessing smoke test and Python compile check passed. The lambda=1 sensitivity rerun reproduced the saved primary metrics to floating-point precision. One stale/mismatched row was found in the saved threshold-calibrated output; a reconciled copy has been generated without overwriting the original.

The current evidence is promising but preliminary. It does not establish that uncertainty-aware selection is universally better or statistically significant. Results vary across datasets and preprocessing configurations, and the search budget is modest.

2. Implementation and protocol

Dataset-level evaluation uses stratified 5-fold outer CV and 5-fold inner CV. The model is an RBF-kernel SVM. PSO searches log10(C) and log10(gamma) in [-3, 3] using 6 particles and 4 iterations. Inner loss is 1 - ROC-AUC. Imputation, scaling, categorical encoding, PCA, and SVD are placed inside the sklearn pipeline, so transformations are fitted within each training partition.

Standard selection: choose the recorded candidate with minimum mean inner-fold loss.
Uncertainty-aware selection: choose the candidate minimizing mean loss + lambda × sample standard deviation of inner-fold loss, with primary lambda = 1.0.

Important scope: PSO particle movement and personal/global best updates optimize mean inner loss for both methods. The uncertainty penalty is applied at the final candidate-selection stage among candidates in the search history. This should be described as uncertainty-aware candidate selection, not uncertainty-aware PSO dynamics.

3. Data and preprocessing

Dataset	Rows	Predictors	Target/preprocessing notes
Breast Cancer Wisconsin (Diagnostic)	569	30	Drop ID; malignant = positive; no missing values reported.
Pima Indians Diabetes	768	8	Zero values in glucose, blood pressure, skin thickness, insulin, BMI are treated as missing; zero p
Cleveland Heart Disease	303	13	Original target 0 vs 1–4 binarized; 6 missing cells in ca/thal; selected categorical features one-hot

The exact provenance and redistribution/license status of the local Pima CSV copy are not yet verified. Record its source URL/version and SHA-256 checksum before publication. Dataset citations are documented in docs/references.md.

4. Validation and consistency checks

Completed checks:

• Preprocessing smoke test passed for baseline, PCA, SVD, and mean-imputation pipelines.
• Python source files compile successfully.
• Recomputed lambda=1 selections from saved candidate histories reproduce the saved primary outer-fold metrics; maximum absolute difference is about 1.7 × 10■¹■.
• The saved calibrated output had one row whose selected C/gamma differed from the primary output (Heart Disease, SVD + median, outer fold 4, Standard). The original files were preserved; a reconciled calibrated copy now uses the same selected candidates as the primary experiment. With those fixed candidate pairs, default and calibrated AUCs match exactly, as expected for a threshold-only change.

Use the reconciled file for aligned calibrated reporting: results/additional_analysis/threshold_calibrated_consistency_checked_outer_folds.csv. The original threshold-calibrated files remain present for auditability.

5. Lambda sensitivity analysis

The sensitivity experiment reselected candidates from each saved PSO candidate pool for lambda ∈ {0, 0.25, 0.5, 1, 2}, then refit the chosen candidate and evaluated the same outer fold. This required 300 outer-fold evaluations across the 3 datasets, 4 preprocessing configurations, 5 folds, and 5 lambda settings. It is conditional on the original candidate pools; PSO was not rerun independently for each lambda.

lambda	Mean ROC-AUC	Mean balanced accuracy	Mean F1	Mean gap
0.00	0.9059	0.7738	0.6408	0.0035
0.25	0.9072	0.7730	0.6402	0.0020
0.50	0.9070	0.7747	0.6425	0.0020
1.00	0.9074	0.7760	0.6501	0.0004
2.00	0.9073	0.7755	0.6477	-0.0003

Across this descriptive pooled summary, lambda=1 has slightly higher mean ROC-AUC and balanced accuracy than lambda=0 and a mean gap closer to zero. Lambda=2 has a slightly lower average gap but does not improve every metric. Because the table pools different datasets and preprocessing pipelines, it is descriptive only; it is not a clinical estimate or a significance test.

6. Threshold calibration

The separate calibration experiment chooses a threshold maximizing balanced accuracy on inner out-of-fold decision scores from the outer-training data, then applies that threshold to scores from a model refitted on all outer-training data. The outer-test labels are not used for threshold selection. There can be score-scale mismatch because calibration scores come from models trained on inner subsets, while the final model is trained on the full outer-training partition. Report calibrated and default-threshold results separately.

7. Current result interpretation

The saved primary and threshold-calibrated results show different patterns by dataset. Breast Cancer AUC is near 0.992–0.993 for both methods, so gains are small. Diabetes shows some AUC improvement under UA but effects on balanced accuracy and F1 depend on preprocessing. Heart Disease shows some balanced-accuracy/F1 gains for selected configurations, but UA does not win every metric. The main defensible conclusion is that the variability penalty can alter candidate choice and sometimes improve out-of-fold performance; whether this is robust requires further testing.

The overtuning gap is defined as outer loss minus the selected candidate's mean inner loss, with loss = 1 - ROC-AUC. A positive gap means outer loss is larger than the inner mean; a negative gap means it is smaller. This gap is diagnostic and does not by itself prove overtuning.

8. Remaining work before submission

1. Verify the exact Pima CSV provenance and license; store source URLs and checksums for all datasets.
2. Expand the search budget or justify the current 6-particle × 4-iteration budget, ideally with a compute-budget comparison.
3. Consider repeated nested CV or additional independent datasets if computationally feasible; five outer folds alone are limited for inferential claims.
4. Add an uncertainty-aware literature review and make the research contribution precise: selection-stage variability penalty over a shared PSO candidate pool.
5. Report mean ± SD and fold-level paired deltas by dataset/configuration. Do not rely only on pooled averages.
6. Compare against the baseline SVM only if its split/preprocessing/evaluation protocol is sufficiently comparable; current `baseline_svm_results.csv` is labeled preliminary.
7. Document environment/package versions, random seeds, dataset checksums, commands, and the exact files used in final tables.
8. Update the manuscript and presentation to distinguish default-threshold from calibrated results and to state limitations.

9. Output files created by this audit

• results/additional_analysis/lambda_sensitivity_outer_folds.csv
• results/additional_analysis/lambda_sensitivity_summary.csv
• results/additional_analysis/lambda_sensitivity_paired_deltas.csv
• results/additional_analysis/lambda1_reproduction_check.csv
• results/additional_analysis/threshold_calibrated_consistency_checked_outer_folds.csv
• results/additional_analysis/threshold_calibrated_consistency_checked_summary.csv
• docs/results_audit.md
• docs/references.md
• updated README.md, docs/evaluation_protocol.md, and docs/dataset_audit.md

Final conclusion

The core experiment is implemented and its saved primary results are reproducible from the recorded candidate histories. The additional sensitivity analysis strengthens the evaluation, but it remains conditional on the original search histories. The project is ready for careful research write-up and discussion, not for a claim that UA is definitively superior or clinically validated.